Sally Khaidem

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Research Interests

Event-Based Vision ▪ Computational Imaging ▪ Neural Rendering ▪ 3D Gaussian Splatting ▪ 3D Scene Understanding ▪ Video Foundation Models ▪ Multimodal Visual Perception

Research Profile

My research focuses on learning-based visual sensing for robust 3D scene understanding. During my PhD, I developed algorithms spanning event-based vision, computational imaging, HDR reconstruction and neural rendering, and more recently extended this work towards Gaussian Splatting, video foundation models and multimodal scene understanding. I am particularly interested in integrating novel sensing modalities with modern 3D vision and embodied perception systems.

Selected Research Highlights

- Developed an event-guided HDR neural rendering framework for motion deblurring and novel view synthesis.
- Proposed efficient lossless compression methods for neuromorphic event streams using novel spatio-temporal representations.
- Designed learning-based methods spanning event-based vision, computational imaging, neural rendering and video scene understanding.

Research and Industrial Experience

IIT Madras & ICMR–National Institute of Epidemiology (ICMR-NIE)

Project Technical Support *Feb 2026 – Present*

Contributing to an interdisciplinary research project on drone-based digital imaging for automated wildlife monitoring. Developing learning-based multimodal computer vision algorithms using synchronized RGB and thermal imagery for robust aerial detection, population counting and behavioural monitoring of Indian Flying Fox colonies.

Samsung Research Institute Bangalore (SRIB)

PrePARe PhD Intern *Jul 2025 – Jan 2026*

Designed and developed a video foundation model for video panoptic segmentation, enabling robust spatio-temporal scene understanding across diverse visual environments.

Selected Publications

Event-based Vision

- **Khaidem, S.**, Kumar, A.N., Sharma, M. and Mitra, K., 2025. **Blur to Brilliance: Neuromorphic Data Guided Deblurring and HDR Novel View Synthesis**, *IEEE Access*.
- **Khaidem, S.**, Jangid, R. and Sharma, M., 2025. **Lightweight and High-Speed Contextual Intensity Image Reconstruction from Event-Based Sensors with Adaptive FastDynamicNet**, *Journal of Electronic Imaging*.
- **Khaidem, S.**, Dharmaraj, A.C. and Sharma, M., 2025. **Towards Efficient Neuromorphic Data Processing: A Novel Representation with Lossless Spatio-Temporal Compression**, *Visual Communications and Image Processing (VCIP), Klagenfurt, Austria*.
- **Khaidem, S.**, Nevatia, A. and Sharma, M., 2023. **A Novel Approach for Neuromorphic Vision Data Compression Based on Deep Belief Network**, *Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP), IIT Ropar*.
- **Khaidem, S.**, Choudhary, R., Sharma, M. and Sankarnarayan, B., 2025. **High Resolution Multi-exposure Stereo Event-Intensity-Depth Database of Natural Scenes**, *IEEE International Conference on Emerging Technologies and Applications (ICETA), IIIT Gwalior*.

Computational Light-Field Imaging

- **Khaidem, S.** and Sharma, M., 2024. **A Deep Belief Network Approach to Scalable Compression of Light Field Data for Auto-Stereoscopic Displays**, *British Machine Vision Conference (BMVC), Glasgow, UK*.

- **Khaidem, S.** and Sharma, M., 2024. **An Integrated Representation & Compression Scheme Based on Convolutional Autoencoders with 4D-DCT Perceptual Encoding for High Dynamic Range Light Fields**, *IEEE International Conference on Information and Communication Technology (CICT), IIIT Allahabad.*
- **Khaidem, S.**, Sharma, M. and Singh, A., 2026. **Hybrid Lightweight Encryption Framework for Light Field Displays Using CNN-Based Multiplicative Layer Encoding and HIGHT-HMAC-SHA256 Scheme**, *Journal of Electronic Imaging* – Accepted.
- Ravishankar, J., **Khaidem, S.** and Sharma, M., 2023. **A Data-Driven Approach Based on Dynamic Mode Decomposition for Efficient Encoding of Dynamic Light Fields**, *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshop, Vancouver, Canada.*
- Ravishankar, J., Sharma, M. and **Khaidem, S.**, 2022. **A Hybrid Tucker-VQ Tensor Sketch Decomposition Model for Coding and Streaming Real World Light Fields Using Stack of Differently Focused Images**, *Pattern Recognition Letters.*
- Ravishankar, J., Sharma, M. and **Khaidem, S.**, 2021. **A Novel Compression Scheme Based on Hybrid Tucker-Vector Quantization via Tensor Sketching for Dynamic Light Fields Acquired Through Coded Aperture Camera**, *International Conference on 3D Immersion (IC3D), Brussels, Belgium.*

Early Research

- **Khaidem, S.**, Keisham, P., Loitongbam, S. and Khumanthem, M., 2020. **Detection and Removal of Impulse Noise from Colour Image Using Lagrange Interpolation and Centre Weighted Vector Median Filter**, *Journal of Advanced Research in Dynamical and Control Systems.*

Selected Research Projects

Learning-Based Event Vision

Mar 2026 – Ongoing: **Video Semantic Segmentation using Event Cameras** — Ongoing
Investigating learning-based methods for event-driven semantic segmentation of dynamic video scenes.

Nov 2024 – Mar 2025: **Towards Efficient Neuromorphic Data Processing** — VCIP 2025
Developed a lossless event compression framework using a voxelized Super Binary Map and Temporal Event Vector with context-adaptive entropy coding.

Jan 2024 – Nov 2024: **NeuReCon: Intensity Image Reconstruction from Event Cameras** — JEI 2025
Developed a context-guided recurrent neural network (FastDynamicNet) for efficient event-to-image reconstruction from asynchronous event streams.

Feb 2023 – Aug 2023: **Neuromorphic Vision Data Compression using Deep Belief Networks** — ICVGIP 2023
Proposed a learning-based lossless compression framework for neuromorphic event streams using spatio-temporal representations and entropy coding.

Neural Rendering & 3D Scene Understanding

Jan 2026 – Ongoing: **Event-Guided Novel View Synthesis and 4D Panoptic Segmentation** — Ongoing
Investigating event-guided NeRF and 3D Gaussian Splatting for novel view synthesis and dynamic scene understanding with panoptic segmentation.

Jul 2025 – Jan 2026: **Video Panoptic Segmentation** — Samsung Research Internship
Developed a VideoMAE-based foundation model for depth estimation and video panoptic segmentation for robust spatio-temporal scene understanding.

Nov 2024 – Jun 2025: **Event-Guided Deblurring and HDR Novel View Synthesis** — IEEE Access 2025
Developed an event-guided framework for image deblurring and HDR novel view synthesis using NeRF and 3D Gaussian Splatting.

Computational Light-Field Imaging

Jul 2024 – Nov 2024: **Scalable Compression for Auto-Stereoscopic Light-Field Displays** — BMVC 2024
Developed a learning-based compression framework for layered light-field displays using Deep Belief Networks and H.265 encoding.

Jul 2024 – Nov 2024: **HDR Light-Field Compression using Convolutional Autoencoders** — CICT 2024
Developed a convolutional autoencoder with 4D-DCT perceptual encoding for efficient HDR light-field compression.

Dec 2022 – Mar 2023: **Dynamic Mode Decomposition for Dynamic Light-Field Encoding** — CVPR Workshop 2023
Developed a dynamic light-field codec using coded-aperture acquisition, Dynamic Mode Decomposition and HEVC

compression.

Jan 2022 – Aug 2022: **Coding and Streaming of Real-World Light Fields using Focal Stacks** PRL 2022
Developed tensor decomposition techniques for efficient coding and streaming of real-world light fields from focal stacks.

Nov 2020 – Mar 2021: **Compression of Dynamic Light Fields using Coded-Aperture Cameras** IC3D 2021
Developed a real-time light-field compression pipeline using tensor sketching, vector quantization and HEVC encoding.

Education

Indian Institute of Technology Madras **Chennai, India**
Ph.D. in Electrical Engineering, CGPA: 7.87/10 *2020 – Present*

National Institute of Technology Manipur **Imphal, India**
M.Tech. in VLSI & Embedded Systems, CGPA: 8.35/10 *2018 – 2020*

BMS College of Engineering **Bengaluru, India**
B.E. in Electronics & Communication Engineering, CGPA: 8.37/10 *2013 – 2017*

Research Technical Expertise

Research Expertise: Event-based Vision, Neural Rendering, 3D Gaussian Splatting, Computational Imaging, HDR Imaging, Light-field Imaging, Scene Understanding, Video Foundation Models

Programming: Python, MATLAB, C/C++

Frameworks: PyTorch, TensorFlow, Keras, OpenCV

Tools: Docker, Git, Blender, Linux, ROS

Honours & Awards

- ANRF International Travel Support (ITS) Grant, 2025
- Institute Half-Time Research Assistantship (HTRA), IIT Madras (2020–2025)
- First Rank, M.Tech. in VLSI & Embedded Systems, National Institute of Technology Manipur
- Qualified Graduate Aptitude Test in Engineering (GATE), Electronics & Communication Engineering, 2019 and 2020

Languages

Manipuri (Native), English (Fluent), Hindi (Fluent), Bengali (Fluent)